



Spatially coherent colour image reconstruction from a trichromatic mosaic with random arrangement of chromatic samples

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Abstract

Recent high resolution imaging of the human retina confirms that the trichromatic cone mosaic follows a random arrangement. Moreover, both the cones' arrangements and proportion widely differ from individual to individual. These findings provide new insights to our understanding of colour vision as most of the previous vision models ignored the mosaic sampling. Here, we propose a cone mosaic sampling simulation applied to colour images. From the simulation, we can infer the processing needs for retrieving spatial and chromatic information from the mosaic without spatial ambiguity. In particular, the focus is on the ability of the visual system to reconstruct coherent spatial information from a plurality of local neighbourhoods. We show that normalized linear processing allows the recovery of achromatic and chromatic information from a mosaic of trichromatic samples arranged randomly. Also, low frequency components of achromatic information can serve to coarsely estimate orientation, which in turn improves the interpolation of chromatic information. An implication for the visual system is the possibility that, in the cortex, the low frequency achromatic spatial information of the magnocellular pathway helps separate chromatic information from the mixed achromatic/chromatic information carried by the parvocellular pathway.

Keywords: colour image processing, colour vision, cone mosaic, midget pathway, random sampling, retinal physiology

Introduction

In many species, including humans, the coding of spatial and chromatic visual information of a daylight scene is performed through a mosaic of cone receptors. At a given instant of time, the image formed by the cone matrix is a patchwork of chromatic responses, measured by each individual cone with either long (L), middle (M) or short (S) wavelength spectral sensitivity. It is important to understand the spatial and chromatic information representations in such an array of chromatic

samples, particularly because the cone types (L, M or S) are randomly arranged in the mosaic, as shown by recent adaptive optics imaging of the retina (Roorda and Williams, 1999). As an example, just imagine that what you are seeing around you is actually sampled through a mosaic composed of a random arrangement of the three different cone types. How can the visual system create a spatially coherent representation of an object's colours and contours from the chromatic mosaiced image composed of several different local chromatic sample patterns?

In this paper, trichromatic mosaic sampling is simulated on colour images. We will deduce how the visual system reconstructs spatial and chromatic information from the cones. The main result is that the visual system should apply processing strategies for extracting spatially coherent information for a plurality of local neighbourhoods due to the random arrangement of

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cone types in the mosaic. We then present the simulation of random chromatic sampling on colour images; present a model for reconstructing spatial and chromatic information from the mosaic; and discuss the implication of the model for the understanding of colour vision and low-level physiology.

It is difficult to experimentally study the mosaic sampling as it is transparent to vision. However, in experimental conditions, desaturated colours appear for a high frequency oblique black and white grating, known as Brewster colour. Williams *et al.* (1993) suggest that this effect is due to trichromatic mosaic sampling. However, the effect is weak and the resolution at which it appears (higher than the cone spacing) suggests that the trichromatic mosaic has no effect on colour and spatial perception (Williams *et al.*, 1991). In opposition, from the physiological point of view, post-receptoral receptive fields are built from different patterns of random arrangement of cones. Hofer *et al.* (2005) have shown that stimulating a single cone with a monochromatic light generates several different sensations. They suggest that these sensations could be due to the local arrangement of the mosaic surrounding the cone (Brainard *et al.*, 2008). However, linking single cone stimulation to a percept, has not provided a clear idea on the processing of the visual system for reconstructing spatially coherent information (Knoblauch and Shevell, 2001). Studies on retinal anatomy and physiology seek to understand how the trichromatic cone mosaic is taken into account in the formation of receptive fields. Despite numerous studies, it remains controversial whether post-receptoral receptive fields emerge from random cone wirings or chromatic specific wirings handling the variability of chromatic neighbourhoods in the retina (Calkins and Sterling, 1999). Until now, there is no anatomical evidence of a specific chromatic wiring (Dacey *et al.*, 1996).

Based on the fact that there is a one-to-one connection from cone to midganglion cell in the parafovea, Paulus and Kröger-Paulus (1983) designed a model for achromatic and chromatic information estimation from a random cone mosaic. Using the same hypothesis, Young and Marrocco (1989) and Lennie *et al.* (1991) found that chromatic receptive fields are strong enough even if they emerge from random chromatic connectivity with their neighbours. But, outside of the fovea, the private one-to-one connection does not persist, challenging the idea of the presence of colour vision in the periphery of the retina (Mullen and Kingdom, 1996; Martin *et al.*, 2001).

Thus, since mosaic sampling is not tractable experimentally, and, despite modelling or physiological studies, there is no consensus on the way achromatic and chromatic information is reconstructed from the cone mosaic. In this paper, we propose a simulation of

random chromatic sampling applied to colour images and a model for reconstruction. From the simulated mosaic one can infer what could be the reconstruction process of the visual system. The model has already been partially described for digital cameras (Alleysson *et al.*, 2005, 2009) and using neural network approaches (Alleysson *et al.*, 2008). Here more details are given for its applicability to colour vision. Contrary to other models, the simulation of colour images provides an objective way to test different hypotheses by evaluating the quality of reconstruction.

The reconstruction should provide an unambiguous representation of achromatic and chromatic information of the scene, as if the image was acquired with trichromatic samples per spatial position. Unambiguous in this context means that achromatic and chromatic spatial information should be preserved through the mosaic/demosaic process, but not that chromatic ambiguity, resulting from metamerism, nor spatial ambiguity resulting from overriding the sampling theorem (i.e. aliasing), is solved by this process. The most critical aspect is to maintain spatial coherence of information from the plurality of chromatic patterns. Consider a network of post-receptor cells that estimates information from the responses of a local part of the cone network in the mosaic. Spatial coherence means that each cell would be able to extract the same achromatic and chromatic information despite a highly variable pattern of chromatic neighbourhoods. We understand that there are differences between our model and the visual system but we think our simulations can help in formulating general rules of what the visual system might do to allow vision from trichromatic mosaic sampling. We will discuss the specific differences that are likely to be important in the Discussion.

Simulation of random chromatic sampling on colour images

Figure 1a shows a colour image defined by three chromatic sensitivities (R, G and B) per spatial location. We claim that this image contains all chromatic information because it is possible to estimate the shade of colour at every spatial position in the RGB colour space. It also contains all the spatial information, as we can estimate a variation of achromatic contrast on a luminance axis at the spatial resolution defined by the spacing between two pixels. We are using this image as the reference. It represents the information content of a visual scene in the brain accounting for colour and spatial vision. *Figure 1b* represents an image from a human retina using adaptive optics (Roorda and Williams, 1999). Red, Green and Blue false colouration indicates the type of cone L, M and S, respectively. Since two different cone types cannot be at the same position

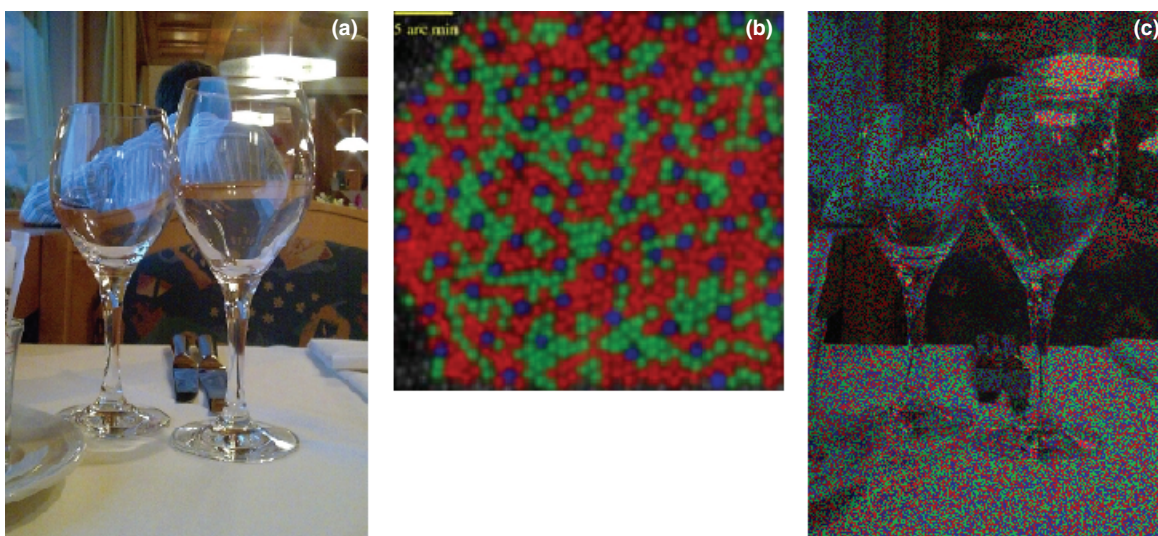


Figure 1. (a) A RGB colour image with three chromatic sensitivities per spatial location. (b) An image of a human retina with cone type (L, M and S) identified by false colouration (R, G and B). (c) Simulation of colour image sampling by a simulated mosaic composed of a random RGB arrangement.

in the retina, they form a mosaic at the surface of the retina. The cone arrangement is mostly hexagonal centred (or equivalently placed on a triangular grid) and the cone types form clusters in the mosaic (Roorda *et al.*, 2001). The simulation of random chromatic sampling applied on a colour image is shown in *Figure 1c*. A complete description of this simulation is available as supporting information (Alleysson, 2010; *Figure S7*; see details provided at the end of the paper). Clearly this image doesn't have all spatial and chromatic information, or at least it is not trivial to extract them from the mosaic representation. We can therefore ask: what is the information representation in such an image? And by analogy with human vision, how is the visual

system able to give us spatially and chromatically unambiguous representations of an object's colours and contours from the cone mosaic?

Information content in a trichromatic mosaic image

A colour image can be decomposed into a spatial achromatic image that contains the spatial information of a scene plus an isoluminant chromatic image, which contains chromatic information. *Figure 2* shows such a decomposition. A colour image is defined as three chromatic values at each spatial position, it is a vector image of three dimensions $\{C_i\}_{i \in \{R,G,B\}}$. These three chromatic values represent the coordinates in RGB

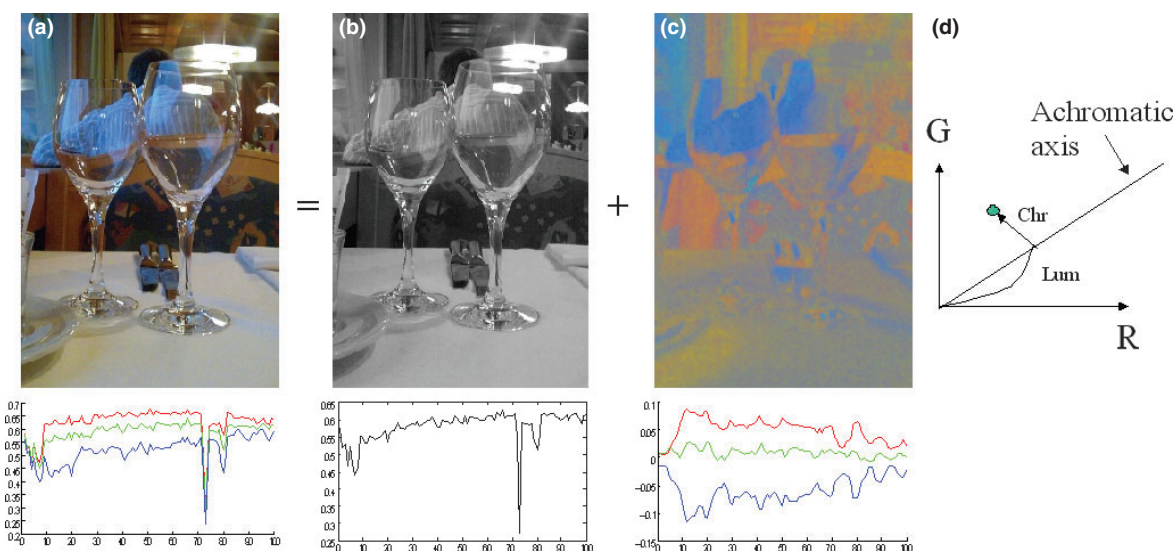


Figure 2. Achromatic (b) and chromatic (c) decomposition of a scene (a). (d) Representation of the decomposition in RG colour space.

colour space at a particular spatial position of a scene. Achromatic information is extracted as a weighted sum of the RGB channels. The values of the weights are determined either to optimise a problem in engineering (Poynton, 2003) or by the nature of human's achromatic coding (Lennie *et al.*, 1993). For now, let them be free and use parameter p_i as the weight. Thus, achromatic information, called luminance, is defined as $L = \sum_i p_i C_i$, with p defining the achromatic axis in RGB space (Figure 2d). Achromatic information has a single value per spatial position, so luminance can be represented with a greyscale image (Figure 2b).

The difference between the colour image and the achromatic image can be calculated by subtracting luminance from each colour channel of the colour image. We thus obtain three opponent chromatic channels (Figure 2c) instead of three chromatic channels, because each channel is subtracted from the weighted sum of the other chromatic channels. We call these channels chrominance and they are defined by $\text{Chr} = \{\text{Chr}_R, \text{Chr}_G, \text{Chr}_B\}$ with, for example, $\text{Chr}_R = R - (p_R.R + p_G.G + p_B.B) = (1-p_R).R + p_G.G + p_B.B$. Figure 2d illustrates the decomposition in RG space.

In a mosaiced image represented in greyscale (because it is cones' responses, see Alleysson, 2010; Figure S7),

due to chromatic subsampling and the loss of position of the cone type, it is impossible to estimate achromatic and chromatic information so simply. But, as a way to understand the information content in a mosaic we can remove the luminance from the mosaic and analyse what remains as information. As shown in Figure 3, if we remove luminance (illustrated in Figure 2b) from the mosaic image (Figure 3a) we get a scalar image (represented in grey scale in Figure 3c). If we demultiplex this image, i.e. we multiply it by three functions m_i (represented in Figure 3d), we obtain an image (Figure 3e) that, after interpolation, reconstructs the chrominance channels (Figure 3f).

This simulation shows that a mosaiced image is actually the sum of achromatic plus chromatic information (Alleysson, 2010; Figure S8). Chromatic information is subsampled and modulated in the mosaiced image, contrary to chromatic information in the colour image. In the frequency domain, as shown in Figure S8 (bottom row), the luminance is coded in the low frequency part of the spectrum, while chrominance is coded in many modulated frequencies due to the random arrangement of chromatic channels in the mosaic. In the simulation discussed above, we did not explain how achromatic information can be estimated

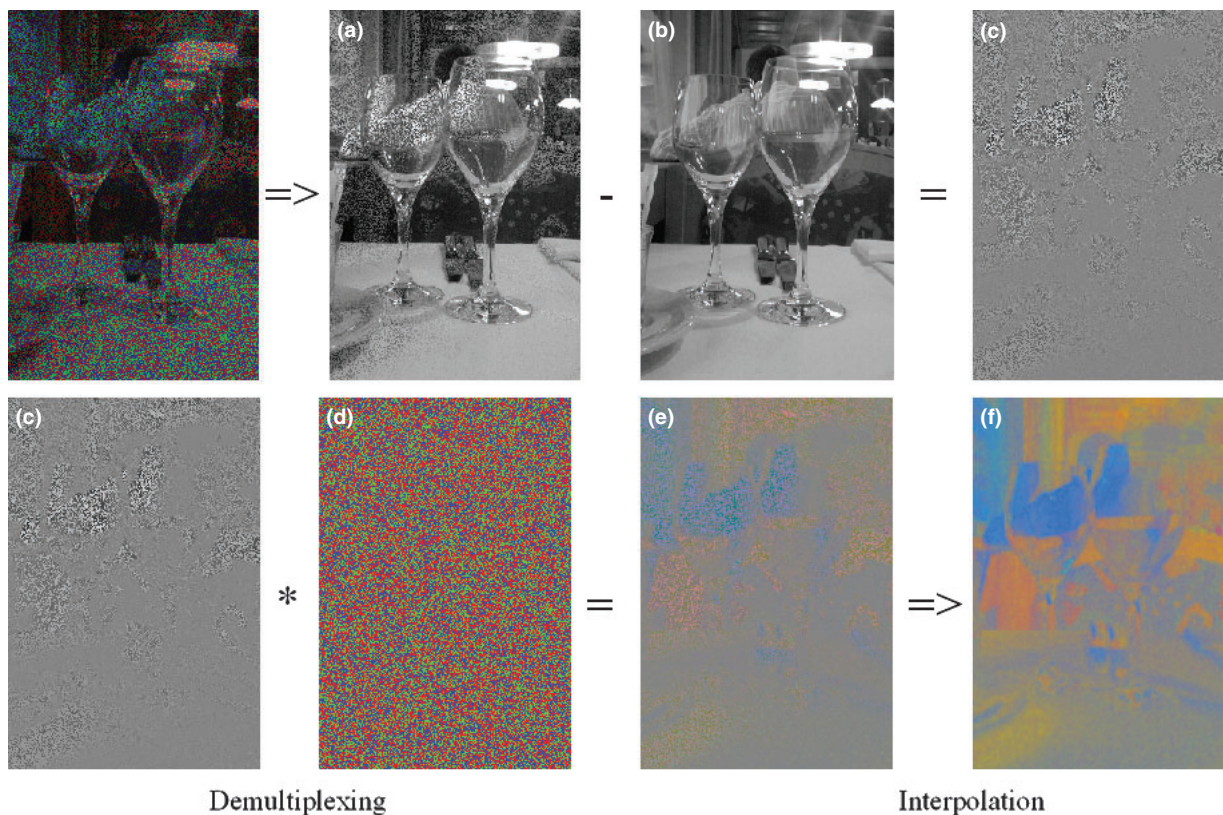


Figure 3. Deciphering information content in a chromatic mosaic with random arrangement of chromatic samples. (a) Image of simulated cone mosaic (b) Achromatic information: luminance (c) Image of the difference $a-b$. (d) Modulation functions m_i (e) Demultiplexed chrominance (f) Interpolated chrominance.

from the mosaic. We do show that the demultiplexing of chromatic information requires a coding of the position for each cone type. Below, we discuss the luminance estimation, the influence of the chromatic topography in the multiplexing and how opponent chromatic information can be interpolated.

Reconstructing images from the mosaic

Luminance estimation

To estimate achromatic information at a position of a chromatic sample in a mosaic, it is necessary to use neighbouring chromatic samples. Each chromatic sample is a sum of achromatic plus specific chromatic information following the particular sensitivity of the receptor at that position. It is not possible to distinguish from a sample's value what the contribution of achromatic vs chromatic information is. Using several samples in the neighbourhood, we can estimate achromatic information from the chromatic samples.

As a first approach, we can think that the luminance should be estimated with a spatially uniform low-pass filter because luminance is coded in the low frequency part of the spatial Fourier transform of the mosaic (Alleysson *et al.*, 2005). However, as illustrated in Figure 4, when using a uniform low-pass filter, the resulting luminance is not constant along the mosaic. The proportion of R, G and B samples in the local neighbourhood defines the luminance axis in the colour space. Since the number of R, G and B samples in a neighbourhood is not constant over the whole mosaic, the luminance has multiple definitions. As a

consequence, the modulation is not completely removed from the mosaic, resulting in a lot of noise in the estimated luminance (Figure 4a).

A non-linear filtering method called normalized convolution is often used for interpolating signals or images that are randomly sampled (Knutsson and Westin, 1993). A derivation of this method can be applied for normalizing luminance estimation. As shown in Figure 4c, we can use the number of neighbours of each type in the neighbourhood to normalise the convolution. This normalisation allows us to estimate achromatic information with a constant weighted p_i (i.e. p_L , p_M , p_S in Figure 4) of the chromatic channel along all the positions in the mosaic. Luminance is then defined uniquely along the mosaic despite a varying number of R, G and B in the neighbourhoods. An example of such an estimation is given in Figure 6b where chrominance channels are marginally interpolated with normalized convolution. This method removes all the modulation noise in the achromatic signal. However, it also removes a lot of high spatial frequency information, so the resulting reconstruction is blurry.

Chrominance demultiplexing

As stated earlier, for recovering three chromatic opponent channels from the mosaic, an operation of demultiplexing should be performed. This operation consists of separating the three different opponent channels from the overall chrominance estimated in the mosaic. In detail, the position of R-L opponent chromatic channels is the position of the R pixels in the mosaic and so on for G-L and B-L at G and B positions. Thus, the demod-

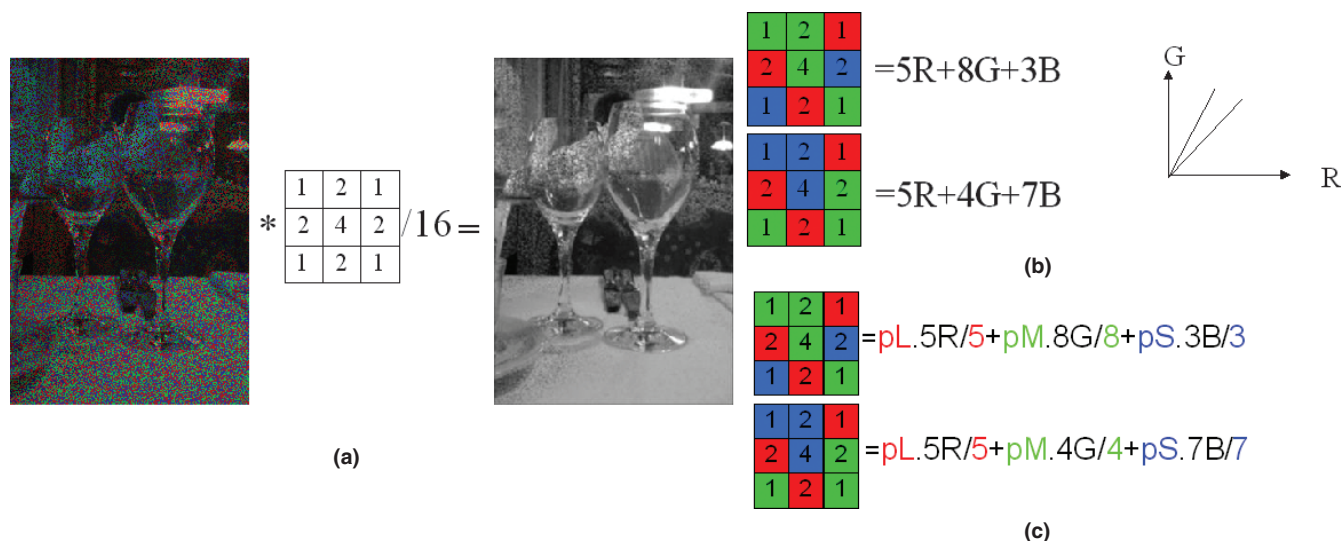


Figure 4. Achromatic estimation with a linear uniform filter (a) Result of the convolution of a uniform filter on the mosaiced image: Noisy reconstruction. (b) Illustration that the random neighbourhood in the convolution kernel of the luminance estimator results in different luminance definitions in RG colour space. (c) Illustration that normalized convolution allows estimating a uniquely defined luminance vector in RG space.

ulation functions are exactly the same as the subsampling functions m_R , m_G and m_B .

To illustrate an incorrect demultiplexing or a poor coding of cone type position we decompose the colour image into its luminance and chrominance. We apply a subsampling of the chrominance according to the mosaic arrangement using m_i functions. Then, we demultiplex the obtained chrominance either by modulation functions m_i or by other random modulation functions. The chromatic information is then interpolated using normalized convolution, on each demultiplexed chrominance. Two colour images were reconstructed that differ only by the demultiplexing operation. We use the sum of the luminance plus either true (*Figure 5a*) or incorrect (*Figure 5b*) demultiplexing followed by chromatic interpolation. This simulation clearly shows that the spatial positions of the cone types needs to be known exactly to accurately recover chromatic information.

Chrominance first

We propose to use the 'Chrominance First' method for the reconstruction of mosaiced colour images. This

method reconstructs chromatic information before achromatic information (Chaix de Lavarenne, 2008; Alleysson *et al.*, 2009). We perform a low frequency estimation of achromatic information from the mosaic using a low pass filter with low cut-off frequency (Alleysson, 2010; *Figure S9b*). This estimation is, of course, partial as only the low frequency information is extracted. Estimating only the low frequencies is advantageous because this frequency band contains less chromatic modulation than the higher frequency bands (Osorio *et al.*, 1998). A more complete description of the method is given in the supporting information (Alleysson, 2010; *Figure S9*) and in the above cited references.

The result of the colour image reconstruction from the mosaic is displayed in *Figure 6*. In *Figure 6b*, we use only normalized convolution for achromatic estimation and chromatic interpolation. In *Figure 6c*, we use the chrominance first framework described in this section and the supporting information (*Figure S9*).

The results show that with chrominance first, spatial information is reconstructed with greater accuracy because the details of the scene are preserved. However,



Figure 5. Illustration of using a wrong demodulation function for demultiplexing the chrominance channels into three opponent chromatic channels. (a) Image reconstruction using a true demultiplexing function (b) Image reconstruction using a wrong demultiplexing function.

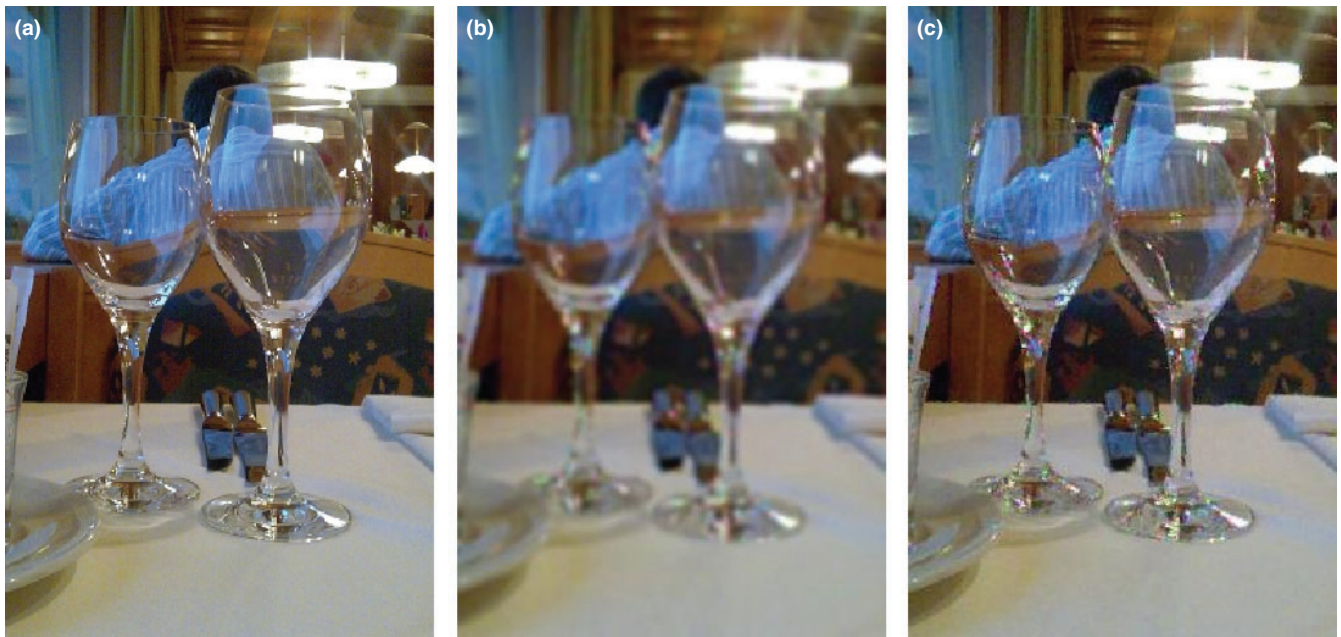


Figure 6. Result of the reconstruction. (a) Original colour image (b) Reconstruction with normalized convolution for achromatic estimation and chromatic interpolation. (c) Reconstruction with the chrominance first method.

some false colour and some blurring remain in the reconstructed image that might be eliminated by considering the temporal aspect of image acquisition. Simulations show that there is no need for a particular treatment of the S-cone photoreceptor. However, the dedicated circuit for S-cones could be a consequence of an evolutionary constraint in colour vision (Regan *et al.*, 2001).

Discussion

By sampling the visual world through a mosaic of cones, the human visual system has to make compromise between spatial and chromatic information. The processing of a trichromatic mosaiced image with random arrangement of chromatic samples allows reconstructing with good accuracy and spatially coherent achromatic and chromatic behaviour. The image processing operations that are required for the reconstruction could be a good illustration of what the visual system should do to provide us with a spatially coherent perception of objects' colours and boundaries.

Models of colour vision

Most colour vision models ignore the sampling by the trichromatic mosaic (Alleysson and Süsstrunk, 2004; Brainard *et al.*, 2008). At a large scale or with large uniform fields, the mosaic is considered transparent for colour vision. Studies about chromatic coding (Buchsbbaum and Gottschalk, 1983) allow understanding of the global behaviour of the visual system for colour vision.

However, ignoring the sampling by the cone mosaic could even lead one to conclude that the retina does not have a role in colour vision, as suggested by Philipona and O'Regan (2006). They show that several colour vision mechanisms could be derived from the singularity of the cone response matrices to natural reflectance; but they do not take into account the fact that the input of the visual system is a trichromatic mosaic rather than a trichromatic stimulus available for each spatial position.

The local arrangement of the chromatic mosaic matters for small object viewing or for studying the relationship between colour perception and retinal physiology. There is, indeed, no consensus on the way the visual system provides humans with a spatially coherent natural scene representation of chromatic and achromatic information from the cone responses arranged in a random mosaic. In Doi *et al.* (2003), a simulated mosaic is used to derive the statistics of the cone mosaic images and to infer post-receptoral processes. They simulated only a small portion of the retina, which is used as a sampling pattern for the whole image. It is possible that this approach works for a different pattern. However, the retinal mosaic requires those patterns to be considered concurrently. This is problematic because those patterns imply different post-receptoral receptive fields. The random nature of the cone mosaic ensures that the neighbourhood of each receptive field is also random. Thus, the tiling of a piece of retina cannot simulate the statistical changes resulting from a different area in the retina. In other words, there is no evidence that the random variable underlying

natural images remains stationary, when considering it as sampled by the random trichromatic arrangement of the cone mosaic (Alleysson and Süssstrunk, 2004). For these reasons, retinal processing could mainly be involved both in providing a spatially coherent representation of the cone responses to natural scenes and in the regulation process improving detection thresholds in the noisy biological system (von der Twer and MacLeod, 2001).

Dendrite tree as a convolution kernel

The function of a dendrite's field that connects one neuron to another through its synapses could be compared to a convolution operator that does a sum of the neighbouring responses weighted by the convolution kernel (De Valois and De Valois, 1980). When neurons are connected to a random mosaic, we may suppose that their dendritic field is spatially variant even if their functions are identical. This is why normalized convolution allows estimating a unique achromatic value along the mosaic despite a different number of specific chromatic samples in its neighbourhood.

By analogy, the horizontal cell layer in the retina which directly connects the cone mosaic, may have dendrite fields where the strength of each dendrite corresponds to the normalized convolution kernel used for estimating low frequency achromatic information. As a consequence, the variability in ganglion cell physiological responses or transfer functions (Lee *et al.*, 2010) would be due to the different arrangement of cones in their receptive field.

Birth of chromatic opponency

We suggest (Alleysson, 2010; *Figure S9a,b*) that the horizontal cells which have a large enough dendritic field (Packer and Dacey, 2005) and do not show chromatic specific wiring (for the H1 subtype) nor chromatic opponency (Dacey *et al.*, 1996) are responsible for the estimation of low-spatial luminance frequency. The opponent R/G chromatic response measured at the midget bipolar cells could then result from the removal of the horizontal cell signals from the cone mosaic (*Figure S9d*). As the mosaic is a sum of achromatic plus chromatic information, a removal of a part of the achromatic signal would enhance the chromatic part. Chromatic opponency that appears at the midget bipolar cell layer could be due to an attenuation of the achromatic part of the signal coming from the cone mosaic. In that case R/G opponency would not result from a chromatic specific mechanism (Calkins and Sterling, 1996, 1999; Dacey, 1999) but from a non-specific normalized achromatic mechanism. This may be

the reason why midget ganglion cell responses show opponency at high retinal eccentricity (Martin *et al.*, 2001) even if these cells' centers receive a pooling of several different cones. Also, this hypothesis is compatible with the fact that this opponency is not strong enough to be perceived psychophysically (Mullen and Kingdom, 1996) contrary to a model of cone specific wiring.

Cone mosaic and magno/parvo pathways

Low spatial frequency information is known to be rapidly conveyed to the primary cortex V1 by the magnocellular pathway. Achromatic high spatial frequency information and chromatic information (R/G), on the other hand, are transmitted together through the midget system by the parvocellular pathway (Ingling and Martinez-Uriegas, 1985; Dacey *et al.*, 2003). It is thus possible that some cells in V1 that respond to orientation (De Valois *et al.*, 1982) help in the interpolation of chromatic information by analysing the orientation of low frequency achromatic information coming from the magnocellular pathway.

The demultiplexing of achromatic and chromatic information coming from the parvocellular pathway is already thought to appear in the primary cortex (Kingdom and Mullen, 1995) and be related to orientation analysis (Martinez-Uriegas, 1993). As an analogy with the digital camera, it is well known that orientation analysis helps in mosaic interpolation (Hamilton and Adams, 1997).

Knowing cone arrangements for demultiplexing

At some level, it is important to know the arrangement of specific cone type positions to be able to recover chromatic information without mixing them. It is not trivial to understand how the arrangement of cone types is coded in the visual system. Wachtler *et al.* (2007) suggest that it is learned from the visual scenes. However, regardless if the position is coded by genetics or learned at the origin of vision, there is a need to reorganise the mosaic in order to separate chromatic specific information. This could be done by projection from the retina to the cortex because those projections may have the property of re-arranging the chromatic components together in the visual cortex.

Packing arrangement of cones in the mosaic

There is evidence that the cone topography does not follow a random assignment and would be rather more packed (Roorda *et al.*, 2001). This could be useful for achromatic estimation because inside a cluster of identical cone types, a variation of the cone's response has a

better chance to be a variation in intensity and not in chromaticity (it is not completely certain because of univariance).

Simulation on a more clustered trichromatic mosaic shows that achromatic estimation is worse when using a packed assignment rather than a uniformly random arrangement (Alleysson, 2010; *Figure S10*). Actually, when facing a cluster, it is harder for the normalized convolution to provide a spatially uniform achromatic signal from the mosaic. Also, interpolating chromatic information would require an increase of the spatial averaging kernel if there are large areas without any of the three chromatic samples present.

It is still not very easy to distinguish between L and M cones, even with adaptive optics, because their spectral sensitivities are too close. A small percentage of identification error between L and M cones could change a random mosaic into a more packed one. But this is still an open question that needs further investigation (Wernet *et al.*, 2006).

Differences between the model and the visual system

One can question whether such a simulation provides a realistic tool for understanding the visual process or if a careful model of the optical path, spectral sensitivity functions, and human mosaics should be included in the simulation (such as the studies using ideal observers (Geisler, 1989; Williams *et al.*, 1993)) as these parameters are presumably different for human cone images and digital camera images. However, it is still unknown whether particular optical, chromatic, and spatial properties of the visual system are used or not to discount the trichromatic mosaic in vision. It is suggested that the conjunction of these properties is finely tuned to allow high spatial acuity and good chromatic discrimination from the cone mosaic (Williams *et al.*, 1993). But, taking into account inter-individual variability such as mosaic topography variability, dichromatism, anomalous trichromatism, and optical modification with ageing, it seems unlikely that a lack of precise conjunction between these variables prevents the visual system from working correctly. Thus, even if digital colour images do not have the same optical, spectral, and spatial properties compared to human cone images, the principle of reconstructing them from the mosaic should not differ qualitatively. They should share a common reconstruction process. It is, of course, possible that the process differs quantitatively at the performance limits, but that remains to be statistically demonstrated using data from several particular individuals, which are not yet available in high enough numbers.

Our simulation is certainly an over-simplification of what could be the real scene acquisition by the human photoreceptors (Hamer and Tyler, 1995; van Hateren,

2007; van Hateren and Snippe, 2007), especially concerning the temporal acquisition (saccade) and processing. Yet, we are convinced that this simplification can help in formulating general principles and we hope that these rules are useful to foster better understanding of the physiology of human colour vision.

Conclusion

The problem of reconstructing full spatial and chromatic information from a mosaic of chromatic samples is not completely solved yet. There remain some artefacts (as illustrated by false colour and blurring) in the method we propose. But, we have focused our research on static colour images, ignoring the temporal aspects of cone image formation in the retina. It is therefore possible that the temporal acquisition and processing in the eye help for the reconstruction (Maloney and Ahumada, 1989).

To be able to understand the precise relationship between the mosaic and the ganglion cell's receptive field, it would be very helpful to measure *in-vivo* both the response of a cell, and the cone arrangement in its receptive field. If this is ever possible we could see clearly what is the local influence of the mosaic on the achromatic and chromatic behaviours on the pre-processing of visual information (Shlens *et al.*, 2009). The simulations and discussion we provide here indicate that the retinal cone mosaic cannot be neglected, as is often done in vision modelling. Indeed, studying its properties and simulating the spatial and chromatic scene reconstruction can provide insights into the functionality of several visual processes.

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Supporting Information

Additional Supporting Information may be found in the online version of this article-<http://dx.doi.org/10.1111/j.1475-1313.2010.00771.x>

Figure S7. The construction of a mosaiced image from a colour image.

Figure S8. Decomposition of a mosaiced image in space and Fourier domain.

Figure S9. Schema of the ‘Chrominance First’ method.

Figure S10. Reconstruction with a packed cone arrangement.

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